

# Mathieu Moonshine Computed, the Double-Scaled Little String Bridge, and Dyons on $K3 \times T^2$ in Lean 4

Xavier Callens<sup>1</sup>  
SocrateAI Scientific Agora Collaboration<sup>1</sup>

<sup>1</sup>*SocrateAI Research Laboratory for Mathematical Physics & Formal Verification*

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## Abstract

We report two Lean 4 libraries (toolchain v4.33.1, Mathlib) in which the mathematics around the  $K3$  elliptic genus is *computed* from closed formulas rather than transcribed from tables. **DualScaleMoonshine** (Stream 4) computes Mathieu moonshine from both sides: the mock modular form  $H^{(2)} = (-2E_2 + 48F_2)/\eta^3$  and its 21 twined series from their formulas, and the traces of all 26 conjugacy classes of  $M_{24}$  from the character table over  $\mathbb{Z}[b_7]$ ,  $\mathbb{Z}[b_{15}]$ ,  $\mathbb{Z}[b_{23}]$ ; the two meet at every class, and the first ten graded pieces of the moonshine module decompose, from computed data, exactly as in Cheng–Duncan–Harvey’s printed table. The same library locates the “24” of the shadow in the polar/finite decomposition of the elliptic genus, and reproduces the arithmetic of Harvey–Murthy–Nazaroglu’s double-scaled little string theories, including a two-sided proof of a divisibility they left unexplained. **DualScaleDyons** (Stream 5) builds the quarter-BPS dyon partition function  $1/\Phi_{10}$  on  $K3 \times T^2$  from the computed elliptic genus, and shows through the orders stated that its single-centred (“immortal”) part at  $m = 1, 2, 3$  is given by Hurwitz class numbers, and that its  $M_{24}$ -twisted versions have virtual-character coefficients. The twining (“forger’s”) test, applied throughout, rejects an integer coincidence proposed earlier in this programme (the “27720 lock”) at all 25 non-identity classes, while  $24 = \chi(K3)$  and the Göttsche numbers pass it. Four reading notes on the literature are recorded. Everything computed is Tier A (kernel-checked; 101 and 44 theorems, 0 failing; axioms `propext`, `Classical.choice`, `Quot.sound` at most); every physical reading is Tier L or Tier C. These are results about the mathematics of  $\mathcal{N} = 4$  string compactifications on  $K3 \times T^2$ , not about our universe: the one experimental prediction of this programme has separately been found excluded by existing data.

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# 1 Introduction

## 1.1 Computed, not typed

Earlier libraries of this programme checked Mathieu moonshine as one typed table against another: the coefficients  $A_1, \dots, A_9$  of Eguchi–Ooguri–Tachikawa [3] against the dimensions of the irreducible representations of  $M_{24}$ . A transcription matching a transcription is weak evidence — and this project has already caught one mistyped  $M_{24}$  table. The two libraries reported here replace transcriptions by computations. The coefficients come from a closed formula; the twined coefficients from the formula for  $H_g$ ; the group-theoretic traces from the character table; the dyon degeneracies from a Borcherds product whose only input is the  $K3$  elliptic genus computed from Jacobi theta functions. Where two independent computations meet, the coincidence is checked by the Lean kernel, not copied. Where a printed table is used, it is used as a *transcription control* that a misreading would break.

## 1.2 The forger’s test

What separated monstrous moonshine from numerology was that the coincidence survives *twining*: replacing a dimension by the trace of every group element. We apply the same test to this programme’s own integers (Section 4). A relation proposed in paper 7 [1] (Theorem 6.1, the “27720 lock”) fails it at every non-identity class;  $24 = \chi(K3)$  and the numbers built from it pass it everywhere they are tested.

## 1.3 Contents

Section 2 restates the tier system. Section 3 computes Mathieu moonshine from both sides. Section 4 applies the forger’s test. Section 5 locates the “24” of the shadow. Section 6 reproduces Harvey–Murthy–Nazaroglu’s double-scaled little strings. Section 7 builds the dyon partition function on  $K3 \times T^2$  and its single-centred part. Section 8 treats the  $M_{24}$ -twisted dyons. Section 9 records four reading notes on the literature. Section 10 places these results against the programme’s experimental verdict. Section 11 states what is and is not proved; Section 12 records methods; Sections 13 and 14 give open problems and research directions; Appendix A is a reproducibility appendix.

# 2 Epistemic tiers

As in papers 7–9, every claim carries one of three tags.

- **Tier A — Lean kernel** — accepted by the Lean 4 kernel: `#print axioms` reports at most `propext`, `Classical.choice`, `Quot.sound`; no `sorry`, no `admit`, no project-declared `axiom`, and no compiled evaluation (`native_decide` is not used anywhere).
- **Tier L — literature** — established in the cited literature, quoted with a file and line range in `papers/foundations/*.txt`.
- **Tier C — conjecture** — a reading or conjecture introduced by this project. **The identification of a Lean object with a physical quantity is never Tier A.**

Nearly all Tier A statements below are finite identities between truncated power series with integer (or Laurent-polynomial) coefficients, decided by kernel evaluation; each theorem states its truncation order. “Through  $q^N$ ” means exactly that and nothing beyond.

### 3 Mathieu moonshine from both sides

#### 3.1 The modular side

Cheng–Harrison [4] (1406\_0619.txt ll. 800–813, eqs. (3.5)–(3.6)) give  $H^{(2)}(\tau) = (-2E_2(\tau) + 48F_2^{(2)}(\tau))/\eta(\tau)^3$  with  $F_2^{(2)} = \sum_{r>s>0, r-s \text{ odd}} (-1)^r s q^{rs/2}$ . Tier L — literature.

**Theorem 3.1** (Computed equals typed). Tier A — Lean kernel *The series computed from this formula, divided exactly by  $\prod(1 - q^n)^3$ , equals  $-2, 2A_1, \dots, 2A_9$  of the EOT table (`hComputed_eq_table`); the transcription of  $F_2$  is checked separately (`f2_control`).*

Cheng–Duncan–Harvey (CDH) [5] give the twined series  $H_g = (\chi_g/24)H + F_g/\eta^3$  (1204\_2779.txt l. 2934, eq. (4.18)) with  $F_g$  the weight-2 forms of their Table 3 (ll. 2882–2915): multiples of  $\Lambda_N$ , eta quotients, and the newforms  $f_{11}, f_{14}, f_{15}, f_{23,a}, f_{23,b}$  with prefactors  $2/5, 1/3, 1/4, 1/3, 1/11$ . Tier L — literature.

**Theorem 3.2** (All 21 columns of CDH Table 20). Tier A — Lean kernel *For every conjugacy class, the twined series computed as  $24D \cdot H_g$  over  $\mathbb{Z}$  equals 24D times the printed column of CDH Table 20 (ll. 4494–4512) through  $q^9$ : `twined_2A`, `twined_3A`, `twined_5A`, `twined_7AB` and `twined_2B`,  $\dots$ , `twined_23AB`; the division by 24D is exact (`twined_div24`, `twinedAll_div`).*

Negative controls: dropping  $F_g$  breaks the match (`twined_2A_needs_F`); dropping the newforms breaks 11A and 23AB (`twined_11A_needs_newform`, `twined_23AB_needs_newforms`); a wrong exponent in the 12A eta quotient breaks it (`twined_12A_exponent_matters`). The two forms that CDH print both as  $\Lambda$ -combinations and as eta quotients agree (`eta_lambda_agree_2B`, `eta_lambda_agree_4A`).

#### 3.2 The group side

CDH Table 8 (ll. 4170–4209) is the character table of  $M_{24}$ , with irrational values  $b_n = (-1 + \sqrt{-n})/2$  for  $n = 7, 15, 23$ . We store  $a + b b_n$  as a pair  $(a, b)$  with  $b_n^2 = -b_n - (n+1)/4$ , and accumulate inner products as a tuple (rational part,  $\omega_7$ -,  $\omega_{15}$ -,  $\omega_{23}$ -part); since  $1, \sqrt{-7}, \sqrt{-15}, \sqrt{-23}$  are  $\mathbb{Q}$ -linearly independent, an inner product equals an integer exactly when the tuple is (integer, 0, 0, 0).

**Theorem 3.3** (Guards on the transcription). Tier A — Lean kernel *The column norms of the transcribed table are the centralizer orders (`colNorm_eq_cent`), each divisible by the element order and dividing  $|M_{24}|$  (`cent_divides`); the class equation holds (`class_equation`); the first orthogonality relation holds for all  $26 \times 26$  pairs of rows, irrational columns included (`gram_ok`); a misreading outside the relabelling orbit violates it (`misread_fails_gram`).*

The flattened PDF text loses the overlines ( $b_7$  and  $\bar{b}_7$  both print as `b7`). An exhaustive search (Python, *not* a kernel proof) found 128 readings passing orthogonality, exactly the orbit under renaming classes and characters, all with the same traces.

**Theorem 3.4** (Moonshine at every class). Tier A — Lean kernel *For all 26 classes  $g$  and levels  $n = 1, \dots, 7$ , the trace of  $g$  on EOT’s representation is a rational integer (the irrational parts vanish) and equals the coefficient of the computed  $H_g$  (`trace_eq_twined_coeff_all`); algebraically conjugate classes have equal traces (`trace_pairs_equal`); swapping two classes breaks the match (`trace_7A_ne_series_23A`).*

#### 3.3 Levels 0–9: the module, from computed data

Reversing the direction, the multiplicity of each irreducible in the graded piece  $K_n$  is  $\langle T_n, \chi_i \rangle$  with  $T_n(g)$  the computed coefficient. CDH print the decompositions in their Table 48 (ll. 5305–5330).

Tier L — literature.

**Theorem 3.5** (The first ten graded pieces). Tier A — Lean kernel For  $n = 0, \dots, 9$  every multiplicity is an integer, equal to CDH Table 48, and non-negative from  $n = 1$  (`moonshine_modules_decompose`, `moonshine_multiplicities_nonneg`). In particular levels 8 and 9, for which EOT give no decomposition, are characters of genuine  $M_{24}$ -modules; e.g.  $K_8 = 990 \oplus 990 \oplus 1035' \oplus 1035'' \oplus 2 \cdot 1265 \oplus \dots \oplus 6 \cdot 10395$ .

Cheng [8] (1005\_5415.txt ll. 540–548) observed that EOT’s original proposal  $30843 = 10395 + 2 \cdot 5796 + 5544 + 3312$  at level 7 is inconsistent with the twined series and proposed  $10395 + 5796 + 5544 + 5313 + 2024 + 1771$ ; the first is refuted by `eot_level7_proposal_inconsistent`, the second is the decomposition used in `trace_eq_twined_coeff_all`. Tier A — Lean kernel.

## 4 The forger’s test

Paper 7’s Theorem 6.1 relates the first two coefficients,  $\mathcal{A}_2/(N_Q \mathcal{A}_1) = 462/360$ , with product 27720. It holds at the identity, now from the computed series (`lock_at_identity`).

**Theorem 4.1** (The lock fails twining). Tier A — Lean kernel The ratio form of the lock, with each twined series’ own coefficients, fails at all 25 non-identity classes (`ratio_fails_at_every_class`; first shown at  $2A, 3A, 5A, 7AB$  by `ratio_fails_at_2A, \dots`). At  $2A$ ,  $\mathcal{A}_2/\mathcal{A}_1 = 14/(-6)$ , not  $462/90$ .

Relations that encode module structure survive twining; this ratio does not. By that criterion the lock is numerology. Tier C — conjecture (the criterion is a reading; the failure is Tier A — Lean kernel).

By contrast, the integers of the dyon sector pass. CDH’s Frame shapes (Table 14, ll. 4349–4364) have degree 24 and fixed points  $\chi_g$  (`frame_degree`, `frame_fixed_points`) and agree with the power maps of Table 8 (`frame_power_maps`); the twined Göttsche numbers  $T_k(g) = [p^k] \prod_{n \geq 1} \det(1 - p^n g | \mathbb{C}^{24})^{-1}$  equal  $p_{24}(k)$  at  $g = 1$  (`twined_goettsche_identity`) and, for  $k \leq 4$ , are characters of  $M_{24}$  with non-negative integer multiplicities (`twined_goettsche_is_character`); e.g.  $H^*(\text{Hilb}^2 K3) = 3 \cdot 1 \oplus 3 \cdot 23 \oplus 252$  (`twined_goettsche_k2`). Tier A — Lean kernel. This is mathematically expected (symmetric powers of a permutation module); its value is the certified cross-check of three transcribed tables and the contrast with the lock.

## 5 Where the 24 of the shadow comes from

Harvey–Murthy–Nazaroglu (HMN) [6] state that the shadow of  $H^{(2)}$  is  $24\eta^3$  (1410\_6174.txt l. 246), and that  $\eta^3 H$  is a mixed mock modular form whose completion is, up to  $-\frac{1}{2}$ , the second helicity supertrace of the double-scaled little string theory at  $k = 2$  (ll. 262–270). Tier L — literature. No completion or modular transformation is formalized here. What is formalized is the arithmetic skeleton, following CDH §2.3 (1204\_2779.txt ll. 657–711, eqs. (2.19)–(2.25)):  $\psi = \Psi_{1,1} Z = \chi \text{Av}^{(2)}[(y+1)/(y-1)] + H \hat{\theta}_1$ , with  $Z = 8(f_2^2 + f_3^2 + f_4^2)$  ((2.49), (2.62), ll. 1266, 1397), cleared of denominators into an identity of  $q$ -series with exact Laurent-polynomial coefficients in  $y$ .

**Theorem 5.1** (The polar/finite decomposition). Tier A — Lean kernel Through  $q^9$ :  $Z(\tau, 0) = 24$  with every higher coefficient zero (`ellipticGenus_z0`);  $Z$  begins  $2y + 20 + 2y^{-1} + q(20y^2 - 128y + 216 - 128y^{-1} + 20y^{-2})$  (`ellipticGenus_first_terms`); the cleared  $\Psi_{1,1}$  reproduces CDH’s printed expansion (`psi11_expansion`); the decomposition holds with  $\chi = 24$  and  $H$  the series computed from  $(-2E_2 + 48F_2)/\eta^3$  (`decomposition`), and fails for  $\chi = 23, 25$  (`decomposition_pins_chi`) and for  $H$  replaced by  $H_{2A}$  (`decomposition_needs_H`); the unary theta series whose multiple is the shadow equals  $\eta^3$  (`shadowTheta_eq_eta3`); CDH’s twined shadow multiplicities are traces on  $1 \oplus 23$  (`shadow_coeff_eq_perm_trace`).

Two independent facts meet at 24: the value  $Z(\tau, 0)$  and the multiplicity of the polar part. That this multiplicity is the shadow’s coefficient is Zwegers’ theorem as recast by CDH. Tier L — literature. Reading the polar (massless) and finite (massive) parts as “macro” and “micro” sectors is Tier C — conjecture.

## 6 Double-scaled little strings

HMN attach to each ADE root system  $Y$  with Coxeter number  $k$  (the lattice of vanishing  $(-2)$ -cycles of  $k$  NS5-branes) a BPS index  $\chi_2^Y = \text{rk}(Y)E_2 - 24F_2^Y$  (1410\_6174.txt l. 732, eq. (2.36)), with  $F_2^{(k,d)}$  an explicit divisor sum (l. 703, eq. (2.29)) combined according to their Table 1, and relate it to umbral moonshine for  $X = Y^{24/\text{rk}(Y)}$  by  $\chi_2^Y = -(\text{rk}(Y)/2)\chi_2^X$  (l. 1266, eq. (4.9)).

Tier L — literature.

**Theorem 6.1** (HMN’s formula, reproduced). Tier A — Lean kernel *The formula reproduces HMN’s printed series for  $Y = A_3$  and  $Y = A_7$  (ll. 1295, 1299; `hmn_A3_printed`, `hmn_A7_printed`); HMN’s  $F_2^{(2,1)}$  equals Cheng–Harrison’s  $F_2$  through  $q^{40}$  (`f2kd_k2_eq_f2Coeff`); and at  $k = 2$ ,  $-2\chi_2^{A_1} = \eta^3 H = -2E_2 + 48F_2$  through  $q^9$  (`hmn_k2_eq_moonshine`).*

**Theorem 6.2** (The umbral relation (4.9)). Tier A — Lean kernel *For  $\ell = 2, 3, 4, 5, 7, 13$  the umbral Jacobi forms  $Z^{(\ell)} = 2\phi_1^{(\ell)}$  built from theta functions (CDH (2.49)–(2.51)) are integral (`umbral_exact`), satisfy  $Z^{(\ell)}(\tau, 0) = 24/(\ell - 1)$  (`umbral_chi`), their polar part removes the pole (`umbral_pole_removed`, `umbral_pole_removed_13`), and HMN’s (4.9) holds against  $Y = A_{\ell-1}$  through  $q^6$  (`hmn_umbral_relation_2`, ..., `hmn_umbral_relation_13`); pairing  $\ell = 3$  with  $A_3$  fails (`hmn_umbral_relation_wrong_Y`).*

HMN observe (ll. 1284–1299) that for umbral  $Y$  all coefficients of  $\chi_2^Y$  are divisible by  $\text{rk}(Y)$ , “but it is not clear why” for the massive states, and that  $A_7$  fails.

**Theorem 6.3** (The divisibility, both directions). Tier A — Lean kernel *For every  $Y$  and every  $n$ ,  $\text{rk}(Y) \mid 24$  implies  $\text{rk}(Y) \mid [q^n]\chi_2^Y$  (`divisible_of_dvd_24`; all coefficients, not a truncation, since  $[q^n]\chi_2^Y = -24(\text{rk}(Y)\sigma(n) + F_2^Y(n))$  with  $F_2^Y(n) \in \mathbb{Z}$ ). Conversely, for every  $A_{k-1}$  ( $k \geq 2$ ) and every  $D_{j+1}$  ( $j \geq 3$ ), all coefficients are divisible by the rank iff the rank divides 24, with the  $q^1$  coefficient as witness (`A_divisible_iff`, `D_divisible_iff`);  $E_6, E_8$  divide and  $E_7$  fails (`E_divisibility`).*

The divisible  $Y$  are exactly HMN’s umbral list. Within HMN’s own formula, the puzzle has an elementary cause: the massive coefficients are  $24\times$  integers. Why the factor in (2.36) is 24 is not addressed; reading it as  $\chi(K3)$  is Tier C — conjecture.

### Closed as not available: the link to $(-2)$ -reflections

The Stream 4 plan asked to relate this counting to the kernel-checked fact that reflections in  $(-2)$ -vectors are lattice isometries (paper 8), or to show that no formal relation is available. In HMN the lattice-theoretic structure enters through the Niemeier lattice  $L_X$  and its Weyl group,  $G^X = \text{Aut}(L_X)/W_X$  (1410\_6174.txt l. 277, eq. (1.10)). Nothing computed above involves a reflection or an isometry;  $\text{rk}(Y)$  enters only as a number. Constructing  $L_{A_1^{24}}$  would not close the gap, which is the isomorphism  $\text{Aut}(L_X)/W_X \cong M_{24}$ ; we record the link as not available rather than implying a closure.

## 7 Dyons on $K3 \times T^2$

### 7.1 The partition function from the elliptic genus

Dabholkar–Murthy–Zagier (DMZ) [7]: quarter-BPS dyons of type II string theory on  $K3 \times T^2$  are counted by  $1/\Phi_{10}(\Omega) = \sum_{m \geq -1} \psi_m(\tau, z) p^m$  (1208\_4074.txt l. 177, eq. (1.1));  $\Phi_{10}$  is the

Borcherds (multiplicative) lift of the  $K3$  elliptic genus  $2\phi_{0,1} = \sum c(4n-l^2)q^n y^l$  (l. 1820, eq. (5.13)). Hence  $\Delta\psi_m = A^{-1}G_{m+1}$  with  $A = \phi_{-2,1}$  and  $\sum_k G_k p^k = \prod_{r \geq 1, s \geq 0, t} (1 - p^r q^s y^t)^{-c(4rs-t^2)}$ .

Tier L — literature .

**Theorem 7.1** (The product, from our  $Z_{K3}$ ). Tier A — Lean kernel *The coefficients of the computed  $Z_{K3}$  depend only on  $4n - l^2$  (`c_depends_only_on_D`); every  $c(D)$  the truncated products read is available through  $q^9$  (`dmvv_reachable`, an enumerated guard, with a failing example `dmvv_unreachable_example`); the integer divisions in the product are exact (`negBinom_exact`). At  $z = 0$ ,  $G_k = p_{24}(k) = 1, 24, 324, 3200, 25650, 176256$  for  $k \leq 5$ , constant in  $\tau$  (`goettsche`, `p24_values`) — the Euler numbers of  $\text{Hilb}^k K3$ , with  $p_{24}$  the 24-coloured partitions (DMZ l. 1070;  $p_{24}(3) = 3200$  at l. 3639).*

**Theorem 7.2** (DMZ (5.16)). Tier A — Lean kernel *The six printed identities  $\Delta\psi_{-1} = A^{-1}, \dots, 72\Delta\psi_4 = 51A^{-1}B^5 + 155E_4AB^3 + 93E_6A^2B^2 + 102E_4^2A^3B + 31E_4E_6A^4$  (`1208_4074.txt` l. 1879–1888), with  $A, B = \phi_{0,1}$  from theta products and  $E_4, E_6$  from divisor sums, hold (`dmz_516_q1`:  $k \leq 5$  through  $q^1$ ; `dmz_516_q2`:  $k \leq 4$  through  $q^2$ ).*

## 7.2 Two-centred and single-centred parts

DMZ’s canonical decomposition  $\psi_m = \psi_m^F + \psi_m^P$  has polar part  $\psi_m^P = p_{24}(m+1)\Delta^{-1}A_{2,m}$ ,  $A_{2,m} = \sum_s q^{ms^2+s} y^{2ms+1} / (1 - q^s y)^2$  (ll. 196–204, eqs. (1.2)–(1.3)), counting two-centred configurations that appear and disappear across walls of marginal stability, while  $\psi_m^F$  counts single-centred (“immortal”) black holes (ll. 424–443). Tier L — literature .  $A_{2,m}$  has different Fourier expansions in different strips; all statements below use  $|q| < |y| < 1$ .

**Theorem 7.3** (The polar part). Tier A — Lean kernel *In the strip, the direct expansion of (1.3) equals DMZ’s (9.55) (l. 5210; `a2m_strip_eq_955`);  $G_{m+1} - p_{24}(m+1)A_{2,m}$  vanishes to second order at  $y = 1$  at every order in  $q$ ,  $m = 1, 2, 3$  (`polar_part_removes_pole`), and  $p_{24}(m+1) \pm 1$  do not (`polar_coefficient_pinned`).*

The second-order condition is automatic by  $y \leftrightarrow y^{-1}$  symmetry; what these two theorems discriminate is the first-order condition — the pole coefficient is  $G_{m+1}(\tau, 0) = p_{24}(m+1)$ . The full second-order content (exact division by  $A$ ) is carried by the theorems below.

**Theorem 7.4** (Immortal dyons are class numbers). Tier A — Lean kernel *With  $H$  the generating function of Hurwitz class numbers, computed by counting reduced binary quadratic forms (`hurwitz_values`), the single-centred counting functions computed from the product satisfy, through the orders stated,*

$$\Delta\psi_1^F = 3E_4A - 648H \quad (q^3), \quad 3\Delta\psi_2^F = -800(12H|V_2) + 22E_4AB - 10E_6A^2 \quad (q^2),$$

$$48\Delta\psi_3^F = -102600(12H|V_3) + 467E_4AB^2 - 430E_6A^2B + 203E_4^2A^3 \quad (q^2),$$

where  $12(H|V_p)$  has coefficients  $12H(\Delta) + p \cdot 12H(\Delta/p^2)$  (`immortal_m1`, `immortal_m2`, `immortal_m3`; exactness `immortal_m1_exact`, `immortal_exact_m23`). Dropping  $H$  breaks  $m = 1$  (`immortal_m1_needs_H`); dropping the Hecke correction breaks  $m = 2$  (`immortal_m2_needs_V2`).

The combinations follow from (5.16) and DMZ’s optimal forms: Example 5 (ll. 3530–3558), (9.10) (l. 3774), (9.11) (l. 3793), (9.12)–(9.13) (ll. 3830, 3850). These are checked, not cited: the finite parts of  $\psi_{0,3}^{\text{opt}}/A$  and  $\psi_{0,4}^{\text{opt}}/A$  are  $-12^{m+1}H|V_m$  (`dmz_911_verified`, `dmz_913_verified_m3`), and DMZ’s printed tables are reproduced (`example5_tables`, `dmz_911_table`, `dmz_912_table`). Tier A — Lean kernel .

## 8 $M_{24}$ -twisted dyons

Cheng [8] proposes twisted denominators  $\Phi_g = e^{4\pi i(\rho, \Omega)} \prod_{\alpha} \exp(-\sum_{k \geq 1} \frac{1}{k} c_{g^k}(|\alpha|^2) e^{2\pi i k(\alpha, \Omega)})$  built from the twisted elliptic genera  $Z_{g^k}$  (1005\_5415.txt ll. 933–942, eqs. (3.9)–(3.10)), with a pole factorizing into two twisted half-BPS functions  $1/\eta_g$  (l. 985, eq. (3.11)); for the geometric classes  $Z_{g^p} = \frac{2}{p+1} \phi_{0,1} + \dots$  (l. 423, eq. (2.5)). [Tier L — literature](#). We compute  $Z_g = (\chi_g/24)Z - F_g A$  for all 26 classes from Stream 4’s data (a consequence of CDH (4.18) and the decomposition of Section 5), and the product by Newton’s identities with CDH’s power maps.

**Theorem 8.1** (Twisted dyons are virtual characters). [Tier A — Lean kernel](#) *The 26 twisted genera are integral, index 1 and satisfy  $Z_g(\tau, 0) = \chi_g$  (`twined_genus_exact`, `twined_genus_index1`, `twined_genus_z0`); Newton’s divisions are exact (`newton_exact`);  $g = 1A$  recovers the untwisted product (`twisted_product_untwisted`); at  $z = 0$  the  $p$ -coefficients are the twined Göttsche numbers (`twisted_goettsche`); every coefficient of  $G_1^{(g)}, G_2^{(g)}$  through  $q^2$  is a virtual character of  $M_{24}$  (`twisted_dyons_are_characters`); for all 26 classes  $T_2(g) A_{2,1}$  removes the double pole (`twisted_polar_removes_pole`); and the twisted single-centred counting function at  $m = 1$  has virtual-character coefficients (`twisted_immortal_are_characters`), e.g. the  $q^1 y^0$  coefficient  $-1800 = 2 \cdot 1 + 2 \cdot 23 + 2 \cdot 45 + 2 \cdot 45 - 231 - 231 + 2 \cdot 252 - 1035' - 1035''$  (`twisted_immortal_example`).*

Once the twisted genera are characters (Gannon’s theorem, [Tier L — literature](#)), the later steps are linear and class-independent, so character-valuedness is expected; the content is the certified end-to-end computation from modular data and CDH’s power maps. That  $1/\Phi_g$  counts twisted dyons, and any  $M_{24}$  action on black-hole microstates, is [Tier L — literature](#) / [Tier C — conjecture](#).

## 9 Reading notes on the literature

These are recorded as observations, not corrections; each is backed by a kernel-checked computation.

1. **HMN (4.2).** 1410\_6174.txt l. 1194 prints  $\chi_2^Y = -\text{rk}(Y)E_2 + 24F_2^Y$ , the opposite overall sign of their (2.36) (l. 732). Their printed series (4.10)–(4.11) (ll. 1295, 1299) and the count of  $\text{rk}(Y)$  massless vector multiplets require the sign of (2.36), which is the one used (`hmn_A3_printed`, `hmn_A7_printed`).
2. **DMZ (1.4).** 1208\_4074.txt l. 211 lists, for the strip  $0 < \text{Im } z < \text{Im } \tau$ , the terms  $\ell q^{(r^2 - \ell^2)/4m} y^r$  with  $r \geq \ell > 0$ . The direct expansion of (1.3) in that strip, and DMZ’s own general formula (9.55), also contain the mirror terms with  $y^{-r}$ ,  $r > \ell > 0$  (from  $s \leq -1$ ); the formalization uses (9.55) (`a2m_strip_eq_955`).
3. **Cheng’s  $c_g(-1)$ .** 1005\_5415.txt l. 960 writes  $c_g(-1) = -2$ . In Cheng’s own (2.5) and in the normalization used here the coefficient of  $y^{\pm 1} q^0$  is  $+2$  for all 26 classes (`twisted_genus_q0`, `c_minus_one`); the printed  $-2$  is a sign convention or a misprint.
4. **EOT’s level-7 decomposition.** Already noted by Cheng (ll. 540–548); confirmed by `eot_level7_proposal_inconsistent`.

## 10 Context: the programme’s experimental verdict

Streams 4 and 5 study the mathematics of  $\mathcal{N} = 4$  string compactifications on  $K3 \times T^2$ ; they make no prediction about our universe. The programme’s single experimental prediction came from Stream 3 (paper 9 [2]): the self-dual length  $s = \sqrt{\ell_P c/H_0} \approx 47 \mu\text{m}$  read as the radius of one large extra dimension. In Stream 6 that prediction was frozen (git tag

`stream6-p1-frozen`; the freeze was made after the bounds were known, and is disclosed as a retrodiction) and compared with the Eöt-Wash 2020 bounds [9] (toroidal radius  $< 30 \mu\text{m}$ , Yukawa range  $< 38.6 \mu\text{m}$ ; 2002\_11761.txt ll. 285–289) and the neutron-star bound of Montero–Vafa–Valenzuela [10] ( $< 44 \mu\text{m}$ ; 2205\_12293.txt l. 327): it fails all three and is excluded by the pre-registered rule (`DualScaleCosmology.Stream6Verdict.p1_verdict_excluded`); the programme’s own T-duality fixes the order-one factor at 1 (`p62_selfdual_fixed_point`), so no convention rescues it. The results of this paper are unaffected: they are Tier A mathematics, and the “micro/macro” readings above were labelled Tier C throughout.

## 11 What is, and is not, proved

### Scope of Lean 4 verification

**Is proved (Tier A; DualScaleMoonshine 101 theorems, DualScaleDyons 44 theorems, 0 failing, 0 sorry; axioms at most propext, Classical.choice, Quot.sound):** finite identities between truncated series, each through the order stated in its theorem —  $H^{(2)}$  and all 21 twined series from their formulas; consistency guards and full orthogonality of the transcribed character table; traces equal computed coefficients at all 26 classes (levels 1–7); multiplicities of  $K_0, \dots, K_9$  equal CDH Table 48; the lock’s failure at 25 classes; the polar/finite decomposition of  $\Psi_{1,1}Z$  with multiplicity 24; HMN’s formula and umbral relation for six lambencies; the divisibility theorem in both directions for all  $A$  and  $D$  types; the dyon product from  $Z_{K3}$ , DMZ’s (5.16), the polar coefficient  $p_{24}(m+1)$ , the immortal parts at  $m = 1, 2, 3$  via class numbers; the twisted dyon products and their character-valued coefficients.

**Is not proved:** mock modularity of any series, any shadow, completion or modular transformation; anything past the stated truncations; the Siegel-modularity of  $\Phi_{10}$  or convergence of the product; that  $1/\Phi_{10}$  or  $1/\Phi_g$  counts dyons, or that  $\psi_m^F$  counts single-centred black holes; the existence of the moonshine module (Gannon); the general statements behind the finite checks (e.g. DMZ’s (9.13) for all primes); that the transcribed character table *is* the character table of  $M_{24}$  (the guards are consistency checks it must pass, not a construction of  $M_{24}$ ); the link to  $(-2)$ -reflections (Section 6); and any statement about our universe.

## 12 Methods

**Exact arithmetic.** Series are lists of integers; Jacobi-form coefficients are Laurent polynomials stored as (offset, coefficient list), with exact convolution — no truncation in  $y$ . Denominators are cleared by hand and the cleared identities are what the kernel checks; exact divisions are asserted by separate theorems. Every formula was prototyped numerically before being stated in Lean.

**Kernel evaluation.** All decisions use `decide`; the heaviest (umbral forms at  $\ell = 13$ , twisted dyons) use `decide +kernel`, in which the kernel evaluates the decision procedure directly — no compiled evaluation and no additional axiom. The HMN file takes about 17 minutes to check, the twisted-dyon file about 8.

**Guards against silent truncation.** A truncated product silently reads unavailable coefficients as zero. The reachability of every coefficient read by the dyon product is an enumerated theorem with a failing example, and every negative control in this paper is a theorem that the corresponding identity is *false*.

**Two errors caught by computation.** An off-by-one in the indexing of  $p_{24}$  (DMZ use the number of 24-coloured partitions of  $n$ , so  $p_{24}(2) = 324$ ) made a first prototype of the polar-part check fail for every coefficient, including the correct one; the failure of the positive case is what

exposed it. A reachability estimate stated as “ $KQ \leq 9$ ” was replaced by an enumeration after review, because the estimate was not connected to the computation it guarded.

## 13 Open problems for the community

1. **Mock modularity in a proof assistant.** No statement here involves a completion or a modular transformation. Formalizing Zwegers’ completion of an Appell–Lerch sum (even for one example) would turn the Tier L step “polar multiplicity = shadow coefficient” into Tier A.
2. **The general theorems behind the finite checks.** DMZ’s (9.13) ( $\Phi_{2,p}^{\text{opt}} = -H|V_p$  for all primes  $p$ ) and HMN’s (4.9) for all umbral  $Y$  are checked here at small order and small index only.
3.  **$D$ - and  $E$ -type umbral forms.** The umbral relation is checked for pure  $A$ -type Niemeier root systems; the  $D$  and  $E$  cases need umbral forms not constructed from the theta products used here.
4. **The reflection bridge.** Formalizing  $\text{Aut}(L_X)/W_X \cong M_{24}$  for  $X = A_1^{24}$ , or any Niemeier lattice, would reopen the link closed as not available in Section 6.
5. **Twisted immortal counts at higher order.** The character-valuedness of the twisted single-centred counts is checked through  $q^2$  at  $m = 1$ ; kernel cost grows quickly with  $m$  and  $q$ .
6. **Why 24 in HMN’s (2.36)?** The divisibility theorem reduces HMN’s puzzle to the factor 24; a derivation of that factor from the DSLST on  $K3$  would complete the explanation.

## 14 Proposed research directions

1. **A library of computed moonshine.** The same method (closed formula, twined formula, character table, kernel comparison) applies to the other umbral groups; CDH’s tables for  $\ell = 3, 4, 5, 7, 13$  are pinned and would serve as transcription controls.
2. **The forger’s test as a standard.** Any integer coincidence proposed as physics should first be twined; formalizing the test for a proposed relation costs little once the twined series exist.
3. **Wall-crossing formalized.** The strip dependence of  $A_{2,m}$  (DMZ (9.55)) is the arithmetic of wall-crossing; checking the jump across a wall against the two-centred degeneracy formula would connect the polar part to its physical interpretation at the level of identities.
4. **CHL models.** For the geometric classes, Cheng’s twisted denominators coincide with CHL dyon counting functions; checking this identification for  $p = 2, 3, 5, 7$  would link the twisted products to an independent physical computation.

## 15 Conclusion

Starting from the  $K3$  elliptic genus computed from theta functions, and from printed tables used only as controls, the kernel confirms Mathieu moonshine at every class and at the first ten levels, locates the 24 of the shadow, reproduces the arithmetic of double-scaled little strings with a two-sided divisibility theorem, and builds the dyon partition function on  $K3 \times T^2$  whose single-centred part at  $m = 1, 2, 3$  is given by class numbers and whose twisted versions are

character-valued. The number  $24 = \chi(K3)$  passes every twining test applied to it; a coincidence proposed earlier in this programme does not. These are results about mathematics, recorded with their scope; the programme’s attempt to extract a prediction about our universe from the same structure has been tested separately and found excluded.

## A Reproducibility

### A.1 Build

```
lake build StringTheoryFoundation DualScaleM24Formalization DoubleFieldTheory \
DualScaleValidation Lean5Corpus StringTheoryFormalization DualScaleStream2 \
DualScaleCosmology DualScaleMoonshine DualScaleDyons
```

All ten libraries build without error (3799 jobs at the Stream 5 release; release tag v3.14.0).

### A.2 Axiom audit

```
python3 tools/axiom_audit.py DualScaleMoonshine
python3 tools/axiom_audit.py DualScaleDyons
```

report **101** and **44 theorems, 0 failing**. The computational theorems depend on `propext` only (six, across the two libraries, on no axiom at all); the divisibility proofs of Section 6 use the three standard axioms. Across the ten libraries of the repository: **611 theorems, 0 failing**.

### A.3 Statement lock

```
python3 tools/statement_lock.py -check $(find DualScaleMoonshine DualScaleDyons -name '*.lean')
```

locks the declarations of both libraries (Stream 4: 9 files; Stream 5: 6 files); every addition was checked to add entries only.

### A.4 Sources pinned

Hash-pinned in `papers/foundations/MANIFEST.md`, line ranges quoted at the point of use: 1004.0956, 1406.0619, 1204.2779, 1410.6174, 1208.4074, 1005.5415, 2002.11761, 2205.12293.

### A.5 Plans

`docs/STREAM4_WORKFLOW.md` and `docs/STREAM5_WORKFLOW.md` record every phase as closed (✓) or closed as not available with the reason; `docs/STREAM6_EXPERIMENT_PLAN.md` records the experimental verdict.

## References

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- [3] T. Eguchi, H. Ooguri, and Y. Tachikawa, “Notes on the K3 Surface and the Mathieu group  $M_{24}$ ,” *Exper. Math.* 20 (2011) 91–96, *arXiv:1004.0956*.
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- [7] A. Dabholkar, S. Murthy, and D. Zagier, “Quantum Black Holes, Wall Crossing, and Mock Modular Forms,” *arXiv:1208.4074* (2012).
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- [10] M. Montero, C. Vafa, and I. Valenzuela, “The Dark Dimension and the Swampland,” *JHEP* 02 (2023) 022, *arXiv:2205.12293*.